Final Report

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Client: Dietrich College Dean's Office of Diversity, Equity, and Inclusion
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Community Partner Background

The mission of the Dietrich College Dean's Office of Diversity, Equity, and Inclusion (DEI) is to "*provide oversight on diversity, equity and inclusion issues and facilitates communication and accountability*" [1]. The office consults an advisory committee and an implementation committee [2], with both parties working together to bring inclusion to community programs, events, and resources. They monitor DEI initiatives within the larger Dietrich ecosystem, which is the second-largest academic body at CMU. Consequently, they reach a large audience that spans across undergraduate students, graduate students, faculty, and staff [3]. Their active goals and objectives are outlined in their Strategic Plan, available on the Dietrich DEI website.

The DEI system relies on shared responsibility, making it decentralized and dynamic. Individual departments launch and monitor their own DEI initiatives, such as workshops, guest lectures, and town hall meetings. These initiatives are organized by part-time volunteers within the department. No enforceable standard existed to tell departments what data to capture, or whether to employ success metrics.

Community Partner Project Opportunity

Project Opportunity

The biggest challenge the Dietrich DEI committee faces is collecting, storing, and accessing committee data in a centralized format. Currently, when events and initiatives are led by committees, there is no defined way for them to forward data such as attendance to our community partner Ayana's office. Ayana often reaches out to the different departments in Dietrich herself to gather the data, which has proven to be extremely inefficient and time-consuming. DEI's structure extends across multiple departments, making it hard to enforce data collection standards.

Establishing a more efficient and organized system for collecting data on DEI initiatives is important because otherwise, the Dietrich College Dean's Office of DEI cannot accurately assess impact. They could not determine how well the programs are doing nor what programs are happening within separate departments. They are missing the opportunity to collect crucial data and to measure the success of all their hard work.

DEI would greatly benefit from a technology solution that allows committees to easily store data, describe their events, and allows our client to instantly view and organize said data. By having a single source for DEI data, tools like Tableau can be utilized to seamlessly create graphics to post to the public. By monitoring trends in the data library, Ayana and the Dietrich College dean would know what works best for implementing DEI efforts in Dietrich College.

Project Vision

Essentially, our proposed project vision was to improve the speed of data collection and data storing within the DEI department.

The biggest pain point that Ayana said she faced was getting a broad picture view of DEI initiatives across Dietrich. It was difficult to find a list of all ongoing DEI events. She had to manually reach out to department heads to ask for relevant post-event data. Such

data is not standardized, not easy to compile/visualize, and is often lost in long email chains. To combat this, we proposed building a standardized form for her to send out to department heads via email. After interviewing stakeholders across Dietrich, we found that our solution addressed four broad pain points: low data-sharing; lack of communication between departments on DEI initiatives; need for improved planning visibility; and motivating volunteers. After filling out the form, the data is automatically stored in a single spreadsheet shared over Google Drive. Using Tableau, our client can develop this into graphics visualizing the state of Dietrich DEI. For example, a pie chart breaking down the number of events by department.

- Value: This solution directly addresses the main pain point that Ayana faces, which is slow communication. Sending a form link, as opposed to taking time to curate questions and then send them over, will reduce the amount of time it takes Ayana to communicate with department heads. Likewise, a pipeline that transfers form data to a spreadsheet will reduce the amount of time it takes for Ayana to compile the data into a place other than her email. A simple Google Drive folder containing the form, responses, and a compiled spreadsheet will allow Ayana to interpret and visualize the data however she wishes to. The main stakeholders for this solution include the department heads, who will benefit from a standardized form with pre-defined data to collect, and Ayana/Dietrich deans, who would now have a centralized area to view data and potentially visualize it.
- Alternate Solution: A possible alternate solution to helping the DEI team with data collection/storage is building a database from scratch, and having department heads enter information directly into it so no emails are sent at all. Though this is probably the most efficient solution for time, it is not feasible for many reasons. Firstly, Ayana emphasized that the department heads are not tech-savvy, and are most comfortable with sending information and communicating via email and Google Suite. Additionally, a custom-built application would require constant maintenance and bug fixes, which would likely be burdensome for Ayana or the CMU IT department.

Project Outcomes

Tools that our team used to develop our technological solution mostly included Tableau and Google Drive. We developed a Google form to Excel spreadsheet pipeline whose responses were available for dynamic Tableau visualizations. See the form and its flowchart in the appendix, figures 5 & 9. The most important feature of this solution was the ability for Ayana to have a centralized space for DEI initiative data. Though we did not develop entirely new technology, we expanded on the ones that Ayana frequently used to communicate with fellow department heads or administrators. Capturing data by form has standardized the data being received, allowing Ayana to compare data points across departments. In addition, the questions will motivate departments to think critically about success metrics and the benefits of implementing data-driven initiatives. The form can be found on the Dietrich DEI website: https://www.cmu.edu/dietrich/about/dei/index.html.

The Dietrich DEI department agreed to use our form to implement a new rewards program, helping motivate volunteers to fill it out. We knew our form would soon fall out of

use without a motivation system. To support this, we brainstormed with our client one person within each department who would be accountable for implementation. Dietrich's senior administrative coordinator will send this list a monthly email, prompting them to fill out the form. Ayana can use this email list to easily communicate with those implementing the form within each department.

The Tableau dashboard has three pages, one shows a summary view of Dietrich, one shows visualizations of the departments and the last shows visualizations regarding each goal. Visualizations include a pie chart broken-down by each DEI initiative's primary objective. Ayana can use these visualizations to identify successful initiatives, departments, or event types. See Appendix B.

In order to have the final outcome of our solution being an official Dietrich data collecting device, we did have to abide by the CMU design guidelines. This included changing the title to include "CMU," making the form dark red, and adding a plaid banner. With regards to how the form was to be distributed to all the DEI departments, our intended outcome was for it to reach as many people as possible. This outcome was achieved as we published the form on the Dietrich DEI official website (https://www.cmu.edu/dietrich/about/dei/index.html). This configuration makes the form accessible to any volunteer or DEI committee member who held an event and would like to relay its information to Ayana. The Tableau visualizations and Google Spreadsheet were only shared with Ayana due to privacy and confidentiality concerns, but this configuration may change as Ayana develops them further after handoff.

In addition to configuring the Google Form aesthetically to fit CMU's image, we conducted several rounds of user testing to ensure that our product was tailored to DEI stakeholders. The user testing expanded Dietrich DEI's capacity by tailoring the form to maximize clarity, efficiency, and value. We asked them about the form's readability and accessibility. One change we made through user testing was adding initiative goal descriptions, so users had a better idea of which bucket their initiative fit into. We had our client validate our initial Google Form before proceeding with development, along with a flowchart visualizing the form's sequence of questions. As the form evolved, we kept her informed by bringing an updated copy to meetings. We user-tested in two phases. In the first phase, we used our first iteration of the form and asked DEI stakeholders to think aloud as they filled out the form for any initiative they have been related to in the past. Similarly, we did the same thing with the second batch of user testers after we altered the form according to feedback. User testing ensured that our product was configured to our user's abilities and expectations.

In addition, we created handover documentation to help the client maintain a sustainable product. The documentation describes the form, how to make edits to it, a system of contact information for people responsible for maintaining the solution, and a list of suggestions for how to improve the form. Our client has read and understands the documentation, improving their capacity to perform these tasks. Another document summarizes our key research findings, documenting all of the key stakeholders and organizations as well as advice on anyone working within Dietrich DEI. We also included a script for testing changes on users, and a video to quickly introduce the form.

We successfully established a solid system of people accountable to our solution. Ayana is the product owner, responsible for changing the form or visualizations if

necessary. We have two people in the IT & Operations department who have agreed to implement Ayana's changes. Each was introduced to the solution. Our client has the capacity to implement changes, and clearly knows who she can ask to do what. When interviewing stakeholders across Dietrich, we also clearly explained the form's purpose over email and in-person. This, as well as introducing the form to the implementation committee, laid the groundwork for implementing the form across all departments.

The main top-level outcome is that Ayana will save countless hours of emailing departments for data and compiling the data they send. The form will no longer have to be emailed to event administrators, as it is on the home page of the DEI website, and their responses will be automatically updated on Ayana's Google Spreadsheet. In addition to this top-level outcome, Ayana's ability to track trends of DEI initiatives can substantially improve the future of Dietrich DEI. Ayana can now have empirical data (and visualizations) as evidence for what events were successful and can have a good understanding of what kinds of events work best for students. Departments also have more of an incentive to hold department-wide DEI initiatives because of the introduction of awards Ayana is distributing. Overall, the top-level outcomes benefit Ayana and the analysis of DEI events.

Final Project Deliverables

- Google Form
  - Final Google Form - "CMU Dietrich DEI Post-Event Data Collection Form" is accessible in the main folder. Any data collected with appear here.
  - Final Google Spreadsheet - "CMU Dietrich DEI Post-Event Data Collection Form" also displays real-time collected data in one space.
  - Previous iterations - Can be found in the "Google Form Iterations" folder.

- Information Flowchart
  - Final Information Flowchart - "Information Flowchart" is accessible in the main folder. It represents the final Google Form in a clean, visual way.
  - Previous iterations - Can be found in the "Information Flowchart Iterations" folder.

- Tableau Dashboard
  - Final Tableau Dashboard - "final_dashboard.twb" is accessible in the Tableau Dashboard folder. Once opened with Tableau, it will display all the dashboards and visualizations when the data source is connected properly.
  - Screenshots - screenshots of the dashboard is accessible in the Tableau Dashboard folder.

- Documentation
  - User Testing Script - "Default User Testing Script" is accessible in the documentation folder.
  - Documentation Guide - "Guide to the Dietrich DEI Post-Event Data Collection Form" is accessible in the documentation folder.
  - Interview Findings - "Research Findings on Dietrich DEI" is accessible in the documentation folder.

- Evidence of Website Integration
    - “Evidence of Website Integration” is accessible in the main folder.
- Pitch Video
    - “pitch_video.mp4” is accessible in the main folder.

Recommendations

Sustaining the solution's capacity depends on maintaining volunteer motivation and ensuring the form stays relevant. Departments and volunteers must be motivated to continue filling out the form. Our client can incentivize them by further pursuing a program that rewards departments for hosting events and then logging them through the form. In addition, our client will have to leverage the system of people accountable to the form in case its frequency of use goes down. In terms of keeping the form relevant, we advise reviewing it after each new strategic plan is released. The form is a great way to support the strategic plan's goals with success metrics.

As included in the handover documentation, the following is a list of recommendations for how to increase DEI's capacity. For one, you could create a post-event survey for event organizers to send out to attendees. The results of this survey could be added as a link in the form. Dietrich DEI could then capture self-reported data from attendees, as well as further gauge the success of events. Success could be linked to metrics derived from the strategic plan. To create this form, a team should repeat our development process. They should 1. interview the DEI dean to determine what data points they need 2. interview various stakeholders including individuals who already sent out post-event surveys 3. have the client validate an initial prototype 4. iterate through user-testing.

Another recommendation would be to create an automated mailer linked to CMU's event calendar. After the event has passed on the calendar, this mailer could automatically email the event coordinator, reminding them to fill out our form. This extension assumes all/most events are uploaded to CMU's event calendar. After interviewing whoever runs CMU's event calendar, a team could develop an automated bot that would email the coordinator. A reminder to fill out the form sent at the right time will improve the percentage of people who do so.

Our final recommendation would be to increase data visibility. Our client could create more visualizations; integrate visualizations into CMU's website; and make data easily accessible to the wider DEI community, allowing them to analyze it for themselves. For example, Tableau could be used to visualize the scorecards found in the strategic plan. These scorecards could be integrated directly into the strategic plan's page. Making more visualizations, and making them more accessible, increases the collected data's value.

To another IS group working within Dietrich DEI, we would advise them on the importance of interviewing a wide range of stakeholders. Dietrich is a broad space, and sometimes two stakeholders had very different opinions on the same subject. Given the number of people within Dietrich, we would also highly recommend rigorous user testing to strengthen your solution. They should also understand that because DEI is volunteer-based, motivation is key and the solution must be compelling.

About the Team

Gloria Moon served as the project manager, leading the team to success with high-quality deliverables. She is a junior majoring in Information Systems and Human-Computer Interaction with a minor in Media Design. She will be a PM intern in Seattle this summer and is pursuing a career in product management.

Neha Choudhari is a junior at Carnegie Mellon pursuing a degree in Information Systems with an additional major in Statistics/Machine Learning. This summer, she will be working in Seattle as a software engineer. In her free time, she enjoys web development and playing the flute.

Gus Saalfeld worked as the documentation manager, synthesizing the team's best practices and research findings to help the client sustain a long-term solution. He is a senior Creative Writing and Information Systems double major. This summer, he is pursuing software engineering and game development positions and will be hiking the Pacific Coast Trail.

Appendix A: Sketches and Work on the Information Flowchart

Figure 1: First Sketch from Ayana

This clarified ideas and the structure of the Google Form in one of our early meetings.

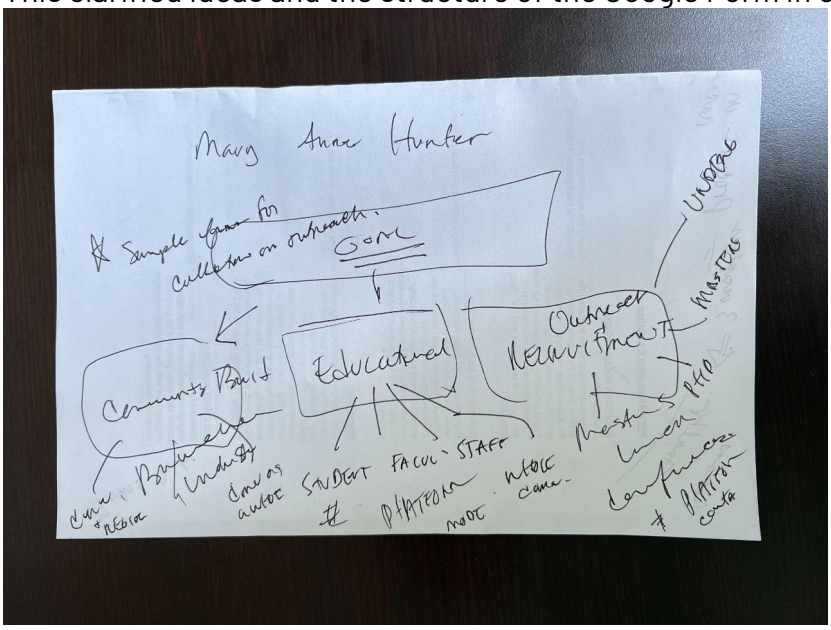


Figure 2: First Sketch from Team

The team then created a sketch based on Ayana's work. This served as a model for the first Google Form.

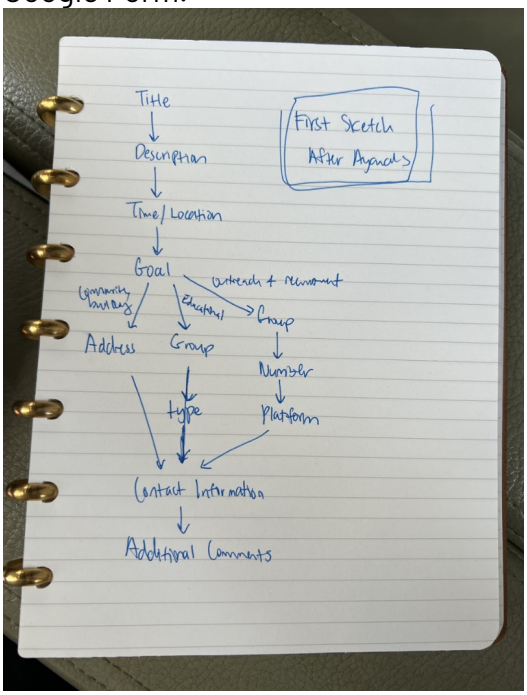


Figure 3: Second Sketch from Ayana

The team printed out the Information Flowchart and reviewed the design with the client in the next client meeting. Ayana commented and suggested areas for improvement.

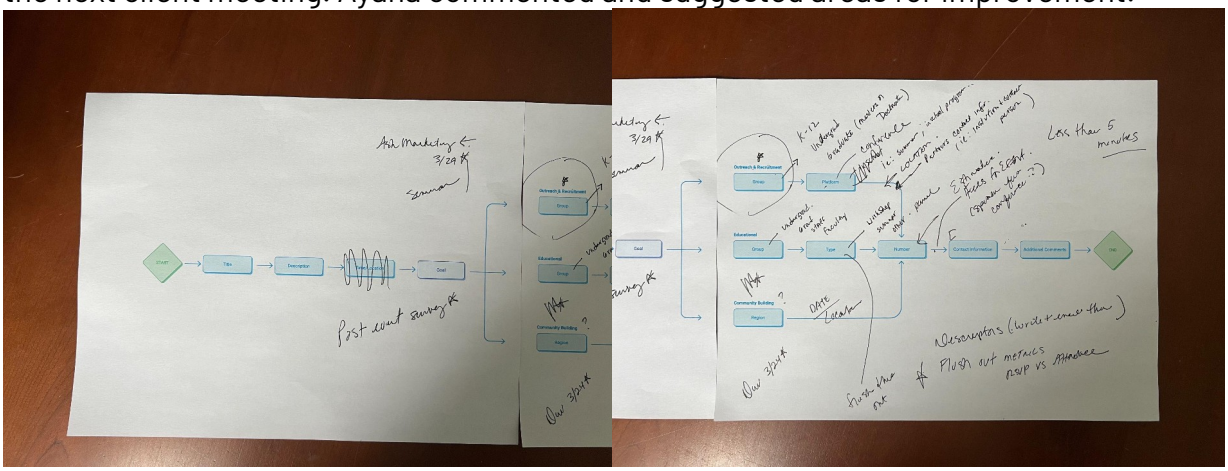


Figure 4: Team Meeting
Reviewed progress on the very last iteration of the Information Flowchart during a meeting.

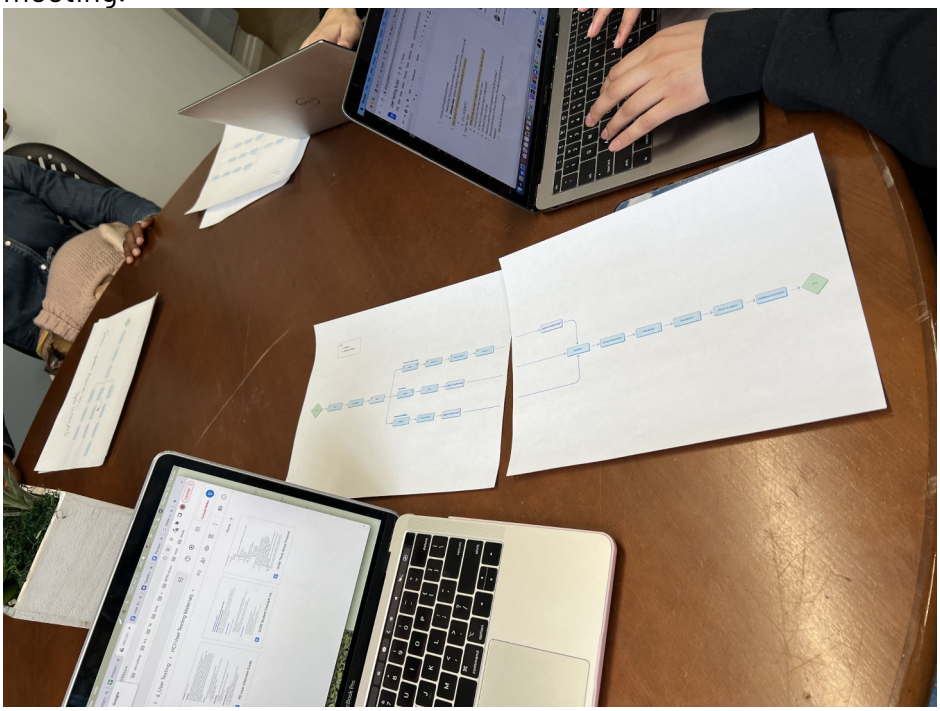
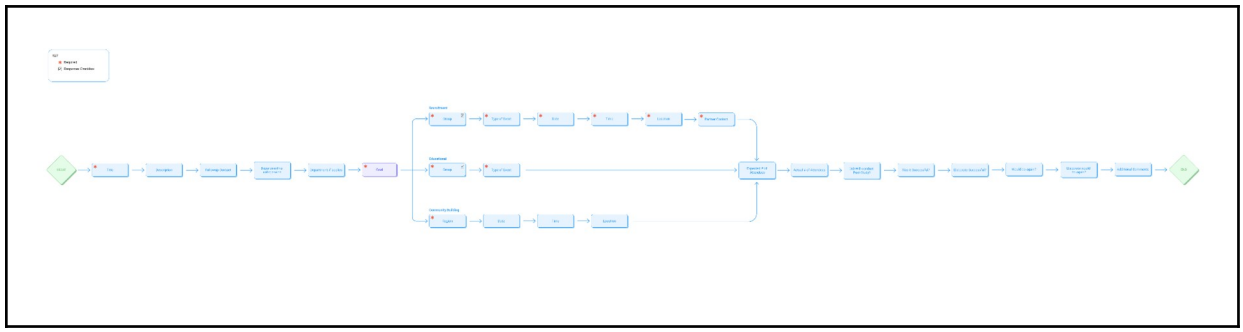


Figure 5: Final Flowchart
The Final Flowchart that visually represents the Google Form solution.



Appendix B: Screenshots of the Tableau Dashboard

Figure 6: Tableau Dashboard 1 - Overall
The main Tableau Dashboard that displays overall information on events and attendance.

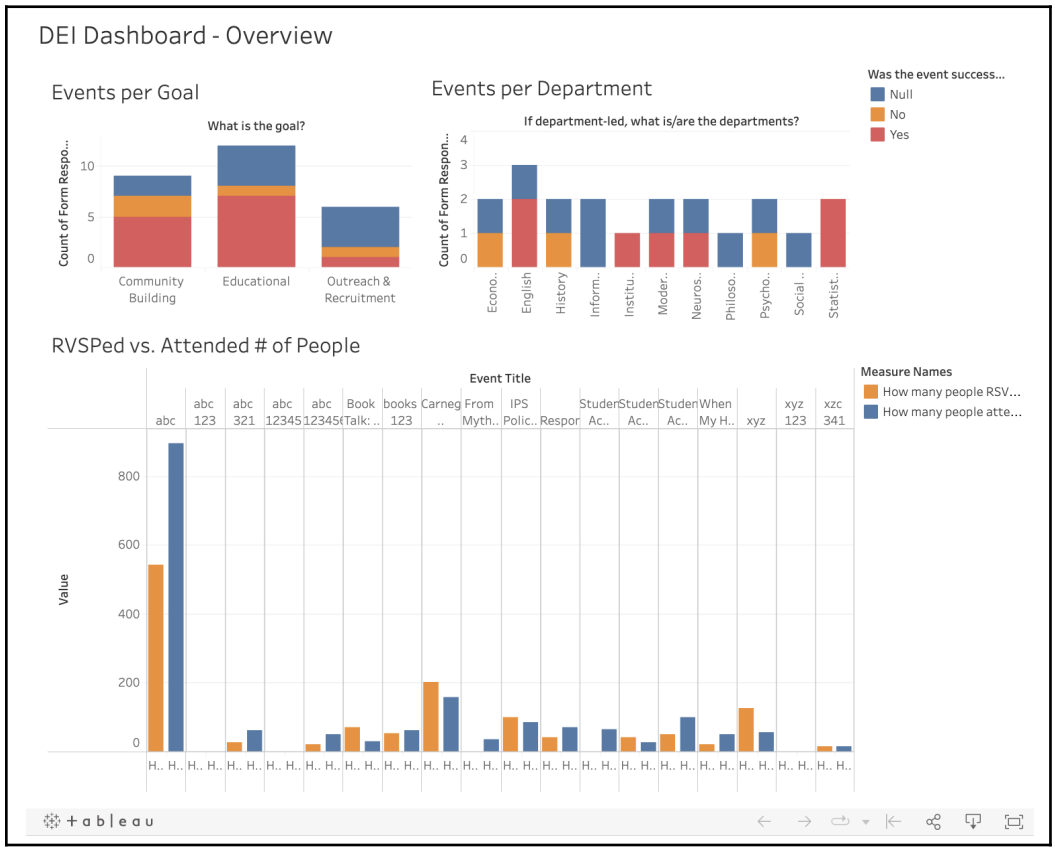


Figure 7: Tableau Dashboard 2 - Departments
The Tableau Dashboard that displays detailed information on departments and events.

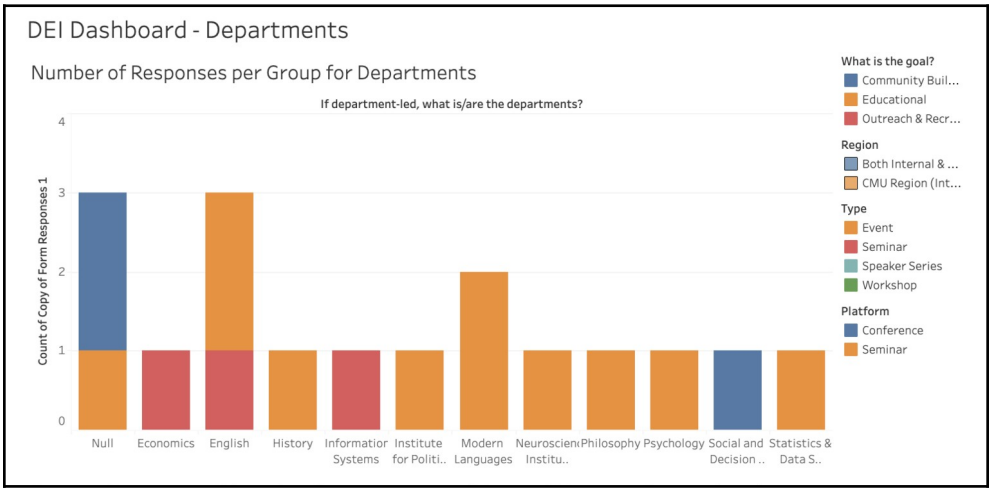
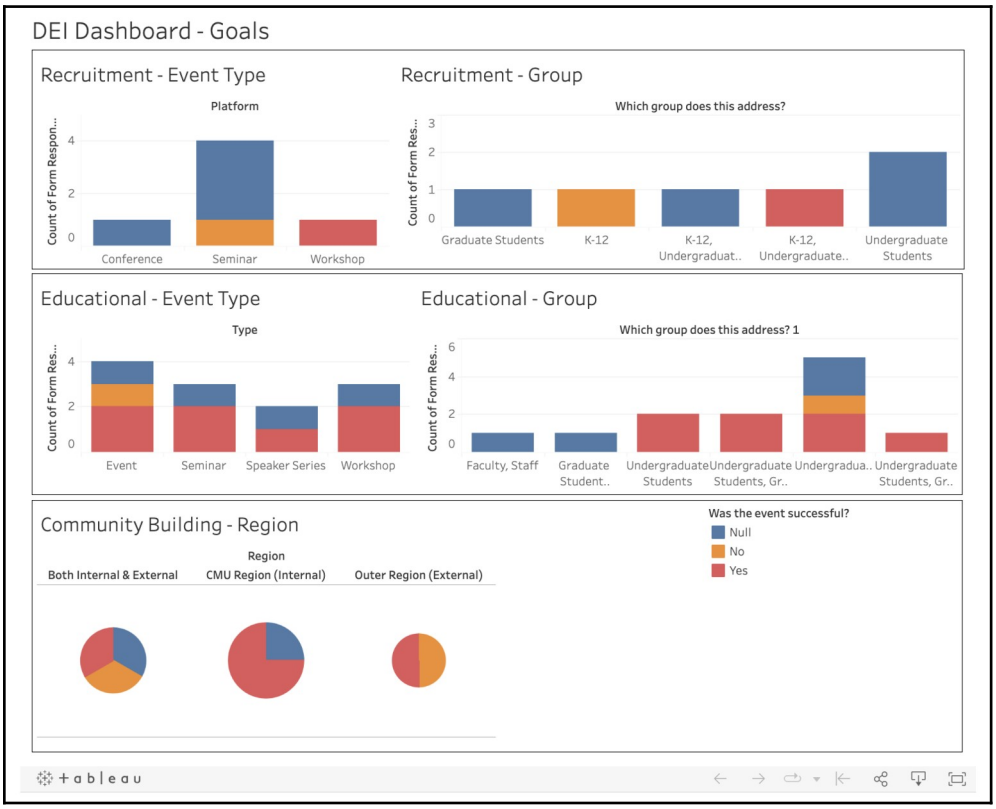


Figure 8: Tableau Dashboard 3: Goals

The Tableau Dashboard that displays detailed information on goals and events.



Appendix C: Google Form

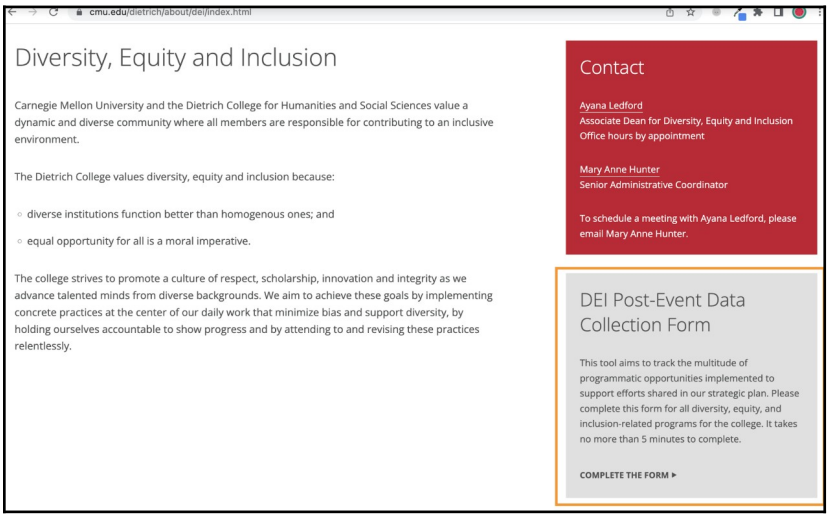
Figure 9: Final Google Form

The Final Google Form solution.



Figure 10: Evidence of Website Integration

A screenshot of the official CMU page where the Google Form is hosted.



Citations

1. https://www.cmu.edu/dietrich/about/dei/index.html
2. https://www.cmu.edu/dietrich/about/dei/committees.html
3. https://www.cmu.edu/dietrich/about/dei/strategic-plan.html
4. https://www.cmu.edu/diehttps://www.cmu.edu/dietrich/about/dei/strategic-plan.htmltrich/about/dei/docs/diversity-equity-inclusion-strategic-plan.pdf